

Summary & contents of the course

Top I: $\pi_n(X, \mathbb{Z}, x)$ and $H_n(X, \mathbb{Z}; A)$

↳ no excision

① Cohomology H^n - contravariant functor

Similar to homology: homotopy invariance, excision

• if $A = k$ - field then

$$H^n(X, \mathbb{Z}; k) \cong \text{Hom}_k(H_n(X, \mathbb{Z}; k), k)$$

• if $\mathbb{Z} = \emptyset$, A - ring then $H^*(X; A)$ - graded ring

$$H^m(X; A) \times H^n(X; A) \xrightarrow{\smile} H^{m+n}(X; A)$$

"cup-product"

Cup product is natural so it is compatible with induced maps f^*

Ex: $H^*(\mathbb{R}P^n; \mathbb{F}_2) \cong \mathbb{F}_2[X]/X^{n+1}$, $x \in H^1(\mathbb{R}P^n, \mathbb{F}_2)$

$$H^*(\mathbb{C}P^n; \mathbb{Z}) \cong \mathbb{Z}[X]/X^{n+1}, \quad x \in H^2(\mathbb{C}P^2; \mathbb{Z})$$

Cor: 1) $\mathbb{C}P^2 \not\cong S^2 \vee S^4$ (homology can't tell this)

2) $\mathbb{Z} \xrightarrow{f} \mathbb{C}P^n$ s.t. $f^* = 0: H^2(\mathbb{C}P^n) \rightarrow H^2(\mathbb{Z})$ then $f^* = 0 \forall \text{ degree} > 0$

Proof: 1) H^2 of the wedge comes from S^2 so it can't generate H^4 which comes from $S^4 \rightarrow 0$. But $x^2 \neq 0$ for some $x \in H^2(\mathbb{C}P^2)$

2) Consider $f^*(x^k) = (f^*(x))^k = 0^k = 0$

□

② Poincaré Duality (top mfd)

[Thm] Let M -compact n -dim mfd, then

$$H^{n-i}(M; \mathbb{F}_2) \cong H_i(M; \mathbb{F}_2)$$

given by "cap-product" with a fundamental class $\mu_M \in H_n(M; \mathbb{F}_2)$

[Corollary] M connected then $H^n(M; \mathbb{F}_2) \cong H_0(M; \mathbb{F}_2) \cong \mathbb{F}_2$
and $H^i(M; \mathbb{F}_2) \times H^{n-i}(M; \mathbb{F}_2) \rightarrow \mathbb{F}_2$

$$(x, y) \longmapsto \varphi(x \cup y)$$

is a non-degenerate bilinear form, i.e.

it induces isomorphism $H^{n-i}(M; \mathbb{F}_2) \rightarrow \text{Hom}(H^i(M; \mathbb{F}_2), \mathbb{F}_2)$

$$y \mapsto \varphi(\cdot \cup y)$$

$$\text{E.g. } H^k(\mathbb{R}P^2; \mathbb{F}_2) = H_{2-k}(\mathbb{R}P^2; \mathbb{F}_2) \cong \begin{cases} \mathbb{F}_2 & k \in \{0, 1, 2\} \\ 0 & \text{else} \end{cases}$$

Let $x \in H^1(\mathbb{R}P^2; \mathbb{F}_2) \cong \mathbb{F}_2$ $\ni x \neq 0$ then

$x \cup x \neq 0$ by the non-degeneracy

$$\Rightarrow H^*(\mathbb{R}P^2; \mathbb{F}_2) \cong \mathbb{F}_2[x]/x^3$$

by bilinearity this computes cup product

With $\mathbb{Z}$ (or $\mathbb{Q}$) coeffs analogous results exist
if M is orientable.

③ Fibrations

Top I: too few tools to compute useful things

$A \subset X$ LES $\pi_*(A, x), \pi_*(X, x), \pi_*(X, A, x)$ but

what $\pi_*(X, A, x)$ even mean?

Def $p: E \rightarrow B$ "Serre fibration" if every

X -CW.
$$\begin{array}{ccc} X \times \{0\} & \xrightarrow{\quad} & E \\ \downarrow \text{co} & \nearrow & \downarrow p \\ X \times I & \xrightarrow{\quad} & B \end{array}$$
 Diagram has lifting that makes triangles commute

Common examples: fibre bundles, covering spaces,

$$S^{2n+1} \rightarrow \mathbb{C}P^n$$

Thm $p: E \rightarrow B$ Serre fibration, $b \in B$, $F = p^{-1}(b)$

$x \in F$ then $\exists$ LES

$$\cdots \rightarrow \pi_n(F, x) \rightarrow \pi_n(E, x) \rightarrow \pi_n(B, x) \rightarrow \pi_{n-1}(F, x) \rightarrow \cdots$$

(better because we have homotopy groups of spaces)

Cor $\pi_k(S^{2n+1}, x) \cong \pi_k(\mathbb{C}P^n, x) \quad k \geq 3$

$n=1$ then $\pi_k(S^3, x) \cong \pi_k(S^2, x) \quad k \geq 3$

Proof: $S^{2n+1} \rightarrow \mathbb{C}P^n$ has fibre S^1 and $\pi_k(S^1, x) = 0 \quad k \geq 2$

$\pi_3(S^2, x) \cong \mathbb{Z}$ w/ generator $\eta: S^3 \rightarrow S^2$

"Hopf map"

④ Mapping spaces

$$\text{map}(X, Y) := \{ f: X \rightarrow Y \text{ cont} \}$$

becomes a space via "compact-open" topology s.t.

Thm If X is locally compact then

$$\text{map}(X, Y) \times X \rightarrow Y$$

$$(f, x) \longmapsto f(x) \quad \text{is continuous}$$

Let Z be a space

$$\{ g: Z \rightarrow \text{map}(X, Y) \text{ cont} \} \longrightarrow \{ g^\wedge: Z \times X \rightarrow Y \text{ cont} \}$$

$$g \longmapsto ((z, x) \mapsto g(z)(x))$$

is a bijection.

Pointed version also exists, and

$$\text{map}_*(S^1, Y) =: \Omega Y \quad \text{"loop space of } Y \text{"}$$

$$\text{Then } \pi_n(\Omega Y, c_{y_0}) \cong \pi_{n+1}(Y, y_0)$$

$\downarrow$
"dual" to suspension

Suspensions are nice wrt (co)homologies
but messy wrt homotopy groups

Loop spaces are nice wrt homotopy groups
but messy wrt (co)homologies.

⑤ Eilenberg-MacLane spaces

$n \geq 1$, A -group ($n > 1$ then abelian)

[Def] An EM space is a path-connected CW complex $(K(A, n), *)$ with an isomorphism $\pi_n(K(A, n), *) \cong A$ s.t.

$$\pi_k(K(A, n), *) = 0 \text{ for } k \neq n.$$

Ex: S^1 is $K(\mathbb{Z}, 1)$, $\mathbb{C}P^\infty$ is $K(\mathbb{Z}, 2)$

[Thm] $K(A, n)$ always exists and unique up to homotopy.

[Thm] Every space X is iteratively built out of EM spaces ↖ "Postnikov tower"

[Thm] $[X, K(A, n)] \cong H^n(X, A)$ is natural

⑥ Steenrod operations

$H^*(X; \mathbb{F}_2)$ has even more natural structure than cup:

$$Sq^i : H^n(X; \mathbb{F}_2) \rightarrow H^{n+i}(X; \mathbb{F}_2) \quad \text{"Steenrod squares"}$$

Together they assemble to a natural action by the Steenrod algebra $\mathcal{A}_2$ on $H^*(X; \mathbb{F}_2)$

Lecture. Cohomology

A -ab group

Recall that homology $H_*(-, A)$:

Spaces $\xrightarrow{p(-)} \mathcal{S}\text{Set} \xrightarrow{C_*(-)} \text{Ch} \xrightarrow{A \otimes -} \text{Ch} \xrightarrow{H_*} \text{graded ab groups}$

Cohomology is the Dual, namely

homology: $A \otimes - \Rightarrow$ cohomology: $\text{Hom}(-, A)$

[Def] A cochain complex C consists of

- ab groups C^n

- homomorphisms $d^n: C^n \rightarrow C^{n+1}$

s.t. $d^{n+1} d^n = 0$

$f: C \rightarrow D$ morphism if $\begin{array}{ccc} C^n & \rightarrow & C^{n+1} \\ f^n \downarrow & \circlearrowright & \downarrow f^{n+1} \\ D^n & \rightarrow & D^{n+1} \end{array}$ for all n

$H^n C = \frac{\ker d^n}{\text{im } d^{n-1}}$ - "n-th cohomology group"

Let $f, g: C \rightarrow D$ morphisms, a cochain homotopy

s b/w f, g (I guess it exists?) satisfies

$\forall n \quad s^n: C^n \rightarrow D^{n-1}$ s.t. $d^{n-1} s^n + s^{n+1} d^n = f^n - g^n$
 $\hookrightarrow$ what are these?

Reindexing $n \mapsto -n$ induces isomorphism of categories

$D: \text{chain cplx} \xrightarrow{\cong} \text{cochain cplx}; D(C)^n = C_{-n}$

$d_{Dc}^n = d_{-n}^c: C_{-n} \rightarrow C_{-n-1}; (Df)^n = f_{-n}$

Then $H^n(DC) = H_{-n}(C)$; $f, g: C \rightarrow C'$ ch homotopic
 $\Leftrightarrow Df, Dg$ are cochain homotopic

Thm $f: C \rightarrow D$ cochain map, $(H^n f)[x] = [f^n(x)]$

Define homomorphisms $H^n f: H^n C \rightarrow H^n D$

- Cochain hom. maps induce the same map in H^n
 $\hookrightarrow$ What does it mean?

- Every SES of cochains $0 \rightarrow A \xrightarrow{f} B \xrightarrow{g} C \rightarrow 0$
gives LES: $\dots \rightarrow H^n A \xrightarrow{H^n f} H^n B \xrightarrow{H^n g} H^n C \xrightarrow{\partial} H^{n+1} A \rightarrow \dots$
with $\partial[x] = [y]$ where $f^{n+1}(y) = d_B^{n+1}(\tilde{x})$ for $g^n(\tilde{x}) = x$

$\text{Hom}(-, A): \text{ch}^{\text{op}} \rightarrow \text{coch}$ (I thought $?$ ch^{op} is cochain)

$\text{Hom}(C, A)^n = \text{Hom}(C_n, A)$; $d^n: \text{Hom}(C, A)^n \xrightarrow{d_{n+1}^*} \text{Hom}(C, A)^{n+1}$
 $f \mapsto f \circ d_{n+1}$

Ex: $\dots \rightarrow 0 \rightarrow 0 \rightarrow \mathbb{Z} \xrightarrow{2} \mathbb{Z} \rightarrow 0 \rightarrow \dots$
 $\downarrow \text{Hom}(-, A)$
 $\dots \leftarrow 0 \leftarrow 0 \leftarrow A \xleftarrow{2} A \leftarrow 0 \leftarrow \dots$

Let s -chain homotopy b/w $f, g: C \rightarrow C'$ then

$\{ \text{Hom}(S_n, A): \text{Hom}(C'_{n+1}, A) \rightarrow \text{Hom}(C_n, A) \}$

is a cochain homotopy of $\text{Hom}(f, A)$ and $\text{Hom}(g, A)$

Def $Y \in sSet$, A - abelian group

The cohomology of Y

$$H^n(Y, A) = H^n(\text{Hom}(C(Y; \mathbb{Z}), A))$$

$Y' \subset Y$ simplicial subset, the relative cohomology

$$H^n(Y, Y'; A) = H^n(\text{Hom}(C(Y; \mathbb{Z})/C(Y'; \mathbb{Z}), A))$$

If X is space, we write

$$\underline{H^n(X; A) = H^n(p(X); A)}$$

$$\text{Hom}(C(Y; \mathbb{Z}), A)^n = \text{Hom}(C_n(Y; \mathbb{Z}), A) =$$

$$= \text{Hom}(\mathbb{Z}[Y_n], A) = \prod_{Y_n} A \quad \text{functions } f: Y_n \rightarrow A$$

("basis" matters only)

$$d^n: \prod_{Y_n} A \rightarrow \prod_{Y_{n+1}} A$$

becomes $d^n(f)(y) = \sum_{i=0}^{n+1} (-1)^i f(d_i^*(y))$

$\swarrow \quad \uparrow$
 $Y_n \rightarrow A \quad Y_{n+1} \quad Y_n$

$Y' \subset Y$ subsimplicial set then

$$\text{Hom}(C_n(Y; \mathbb{Z})/C_n(Y'; \mathbb{Z}), A) = \text{Hom}(\mathbb{Z}[Y_n]/\mathbb{Z}[Y'_n], A) =$$

$$= \{ f: Y_n \rightarrow A \text{ s.t. } f(y') = 0 \}$$

Which means $\exists$ SES

restriction map
 $\downarrow$

$$0 \rightarrow \text{Hom}(C(Y; \mathbb{Z})/C(Y'; \mathbb{Z}), A) \rightarrow \text{Hom}(C(Y; \mathbb{Z}), A) \rightarrow \text{Hom}(C(Y'; \mathbb{Z}), A) \rightarrow 0$$

by noticing

$$0 \rightarrow \{ f: Y_n \rightarrow A, f(y'_n) = 0 \} \rightarrow \{ f: Y_n \rightarrow A \} \rightarrow \{ f: Y'_n \rightarrow A \} \rightarrow 0$$

Key properties

1) Homotopy invariance: $f \simeq g: X \rightarrow Y \Rightarrow H^n(f, A) = H^n(g, A)$

Proof: $C(f; \mathbb{Z}), C(g; \mathbb{Z})$ are chain homotopic

$\Rightarrow \text{Hom}(C(f; \mathbb{Z}), A), \text{Hom}(C(g; \mathbb{Z}), A)$ are

cochain homotopic $\Rightarrow$ induce the same map

2) LES $Y' \subset Y$ subsimp set then

i) $\exists$ SES $\Rightarrow$ ii) induces LES

3) Excision: $\mathcal{O} = \{O_i\}_{i \in I}$ - admissible cover

i.e. $X = \bigcup_I O_i$

Recall (I don't remember) $C(p_{\mathcal{O}}(X); \mathbb{Z}) \xrightarrow{i} C(p(X), \mathbb{Z})$

is a quasi-isomorphism, $p_{\mathcal{O}}(X) \subset p(X)$

is the subsimp set of $\mathcal{O}$ -small simplices

(i.e. $\Delta^n \rightarrow X$ is contained in one of the O_i 's)

Ideally, we want

$$\text{Hom}(-, A) : \text{Hom}(C(p(X); \mathbb{Z}), A) \rightarrow \text{Hom}(C(p_{\mathcal{O}}(X); \mathbb{Z}), A)$$

to be qis (so that it induces iso on H^n)

Warning It is generally not the case

that $\text{Hom}(-, A)$ preserve qis

Basic counterexample:

$$0 \rightarrow \mathbb{Z} \xrightarrow{2} \mathbb{Z} \rightarrow 0$$

$$0 \rightarrow 0 \rightarrow \mathbb{Z}/2 \rightarrow 0$$

$\downarrow f$

are qis but

$$0 \leftarrow \mathbb{Z} \xleftarrow{2} \mathbb{Z} \leftarrow 0$$

$$0 \leftarrow 0 \leftarrow 0 \leftarrow 0$$

$\uparrow f^*$

are not qis

Are we doomed? No

[Thm] (Whitehead thm for chains)

Let $\delta: C \rightarrow D$ be qis, if C, D are bounded below and degree-wise free (more generally, projective)

then δ is a chain homotopy equivalence

Note that $C(p_0(X); \mathbb{Z}) \xrightarrow{\sim} C(p(X); \mathbb{Z})$

is degree-wise free so it is a ch homotopy equiv.

$\text{Hom}(-, A)$ sends ch homotopies to cochain homotopies
so it induces iso on H^*

If (X, Y, U) is an excisive triple: $U \subset Y \subset X$, $\bar{U} \subset \mathring{Y}$

we can use that $\text{Hom}(-, A)$ is gis for the admissible cover $X = (X \setminus U) \cup Y$

the homological excision proof can be dualised.

$$\boxed{\text{Thm}} \quad (X \setminus U, Y \setminus U) \hookrightarrow (X, Y)$$

induces isomorphism

$$H(X, Y; A) \xrightarrow{\cong} H^n(X \setminus U, Y \setminus U; A)$$

$\boxed{\text{Cor}}$ (X, Y) pair that satisfies HEP then

$$H^n(X/Y, Y/Y; A) \xrightarrow{\cong} H^n(X, Y; A)$$

$$H^m(S^n; A) = \begin{cases} A \oplus A & m=n=0 \\ A & n>0 \quad m=0 \text{ or } m=n \\ 0 & \text{else} \end{cases}$$

$$H^m(D^n, S^{n-1}; A) = \begin{cases} A & m=n \\ 0 & \text{else} \end{cases}$$

Cellular cochain X-CW

$$C_{\text{cell}}^0(X; A) := \text{Hom}(C_{\bullet}^{\text{cell}}(X; \mathbb{Z}), A)$$

Recall: $C_n^{\text{cell}}(X; \mathbb{Z}) = H_n(X_n, X_{n-1}; \mathbb{Z})$ so

$$C_{\text{cell}}^n(X; A) \cong H^n(X_n, X_{n-1}; A) \quad (\text{proof is later})$$

$$\boxed{\text{Thm}} \quad H^n(C_{\text{cell}}^*(X; A)) \cong H^n(X; A) \quad \text{naturally}$$